American International University Bangladesh

Computer Graphics

Course Code: CSC4118

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Lecturer

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TITLE: WASHINGTON CITY VIEW

Group Member Name	ID	Contribution
1. Jayed Hasan (Corresponding member)	22-46364-1	Day/Night Scene Development, River and Weather Effects, Animations (Cars, Boats, Thunder, Sun, Moon), User Interaction Features
2. Dipra Kar	22-46336-1	Monsoon and Winter Scene Development, Animations (Rain, Snowfall, Clouds), Building and Bridge Design
Section:	**G**	

Introduction

The Washington City View project uses OpenGL to create a realistic city environment in a virtual simulation of Washington, DC. With interactive and animated features like moving cars, boats, and weather effects like rain and snowfall, the project offers four distinct scenarios: day, night, monsoon, and winter. The goal of this project is to demonstrate how computer graphics can be used to produce realistic simulations for entertainment and urban visualization. Through the integration of technical accuracy and imaginative design, the project showcases the capabilities of modern graphical tools and methods.

Environment and Methods

This project has been developed using:

- **Environment**: OpenGL library in the C++ programming language, integrated with the GLUT toolkit. The development is carried out using the Code Blocks IDE on Windows OS.
- **Methods and Tools**: Geometric primitives, vector graphics, and advanced OpenGL functionalities like transformations and color blending. The code is modularized into functions for ease of design and maintenance.

Feature Set

The project includes the following features:

1. Dynamic day and night transitions. (Jayed)

2. Animated moving clouds with realistic wind effects. (Dipra)

3. Lifelike river rendering with distinct day and night variations. (Jayed)

4. Seasonal views: Monsoon (rain effects) and Winter (snowfall effects). (Dipra)

5. Detailed city skyline with vector-based buildings and distinct textures for day and night. (Dipra)

6. Moving vehicles (cars) with headlights at night on a detailed Washington Bridge. (Jayed)
7. Boats navigating the river with lights during the night view. (Jayed)
8. Realistic weather effects like thunder and rainfall during monsoon. (Jayed)
9. Sun and moon rendering based on the time of day, along with starry night skies. (Jayed)
10. Gradual visual gradients for smooth sky transitions during sunrise and sunset. (Jayed)
11. Snowfall animations with varying sizes and intensities for winter scenes. (Dipra)
12. Modular architecture for adding and extending features in the future. (Jayed)
13. OpenGL transformations for smooth and realistic object movements. (Dipra)
14. Interactive keyboard controls for scenario switching and animations. (Jayed)
15. Shadows and lighting effects for depth perception during day and night. (Dipra)
16. Flowing water effects in rivers with reflections during day and night. (Jayed)
17. Minimal lag for real-time rendering and efficient memory utilization. (Dipra)
18. Pause/resume animation feature for interactive exploration. (Jayed)
19. Adjustable camera angles for enhanced views and usability. (Jayed)
20. Seamless transitions between different weather and time scenarios. (Dipra)

Conclusion

The Washington City View project is a prime example of the technical and artistic potential of computer graphics. With its dynamic weather, day-night transitions, and captivating animations, it accurately represents Washington, DC. Future improvements, like the addition of new landmarks or the integration of real-time data for adaptive weather patterns, are made possible by the project's modularity. Despite drawbacks such as hardware dependence, the project offers a strong foundation for investigating the possibilities of graphical simulations in the fields of entertainment, education, and urban planning. The team's completion of this project has given us invaluable experience using OpenGL for practical purposes.